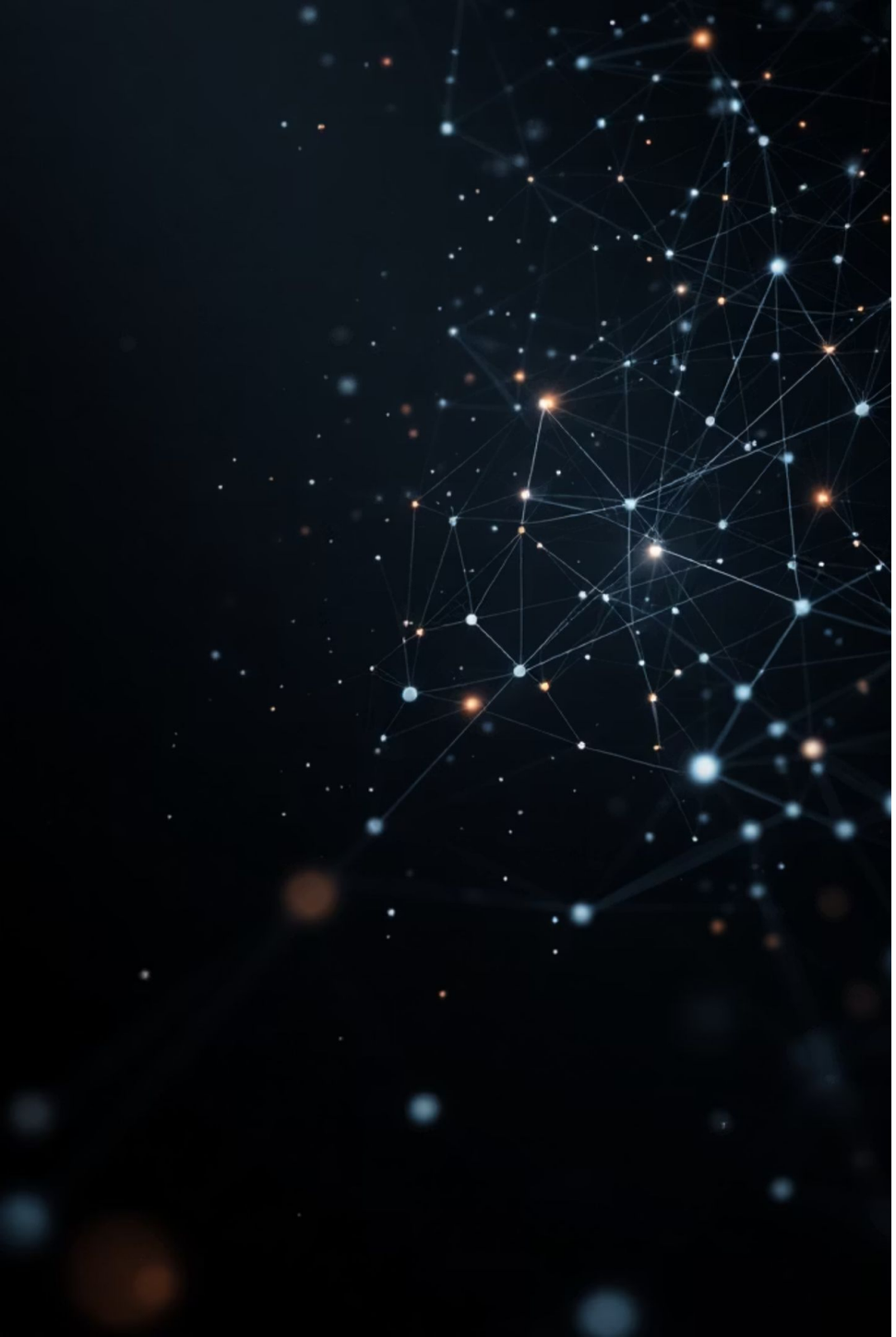
3.3.2 Performance and Cost Monitoring

A look at how AI systems are monitored for performance efficiency and cost management covering prompt compression, workload analysis, storage policies, and cost-benefit analysis.



What is Performance Monitoring?

Core Purpose

Performance monitoring ensures the system can meet workload demand and is properly configured for optimized operation. Understanding how the system responds to heavy workloads and processes queries in a timely manner is essential.

Prompts

User-provided inputs of what they want from the AI system — the starting point of every interaction.

Queries

The backend structured requests processed by the system to generate an output for the user.

Workload Response

How the system handles heavy loads and processes queries in a timely, efficient manner.

Accuracy and Response Quality

Checking for accuracy in responses ensures that the model has not suffered from quality issues. Monitoring the volume of requests and responses is a key indicator of system health.

Monitor Accuracy

Regularly verify that responses are accurate and that the model has not degraded in quality over time.

Track Request Volume

Performance should be monitored by looking at the number of requests and responses being submitted and supplied by the system.

Reduce Supplemental Queries

If every initial response requires supplemental queries to reach a satisfactory answer, the organization may need to improve response detail and specificity.

Prompt Compression

Prompt compression reduces the complexity of a prompt while preserving its meaning and required information. In token-limited environments, this technique is essential for managing both cost and speed.

→ Preserve Meaning

Reduce complexity without losing the required information or intent of the original prompt.

→ Lower Token Cost

Necessary to reduce cost in a token-limited environment — fewer tokens means lower billing.

→ Speed Up Inference

Assists in speeding up inference and reducing cost since the workload to arrive at an output is lower.

Why It Matters

In high-volume AI deployments, even small reductions in prompt length compound into significant cost savings and faster response times across thousands of daily interactions.

Monitoring Requests and Reducing Query Volume

Drive Model Improvements

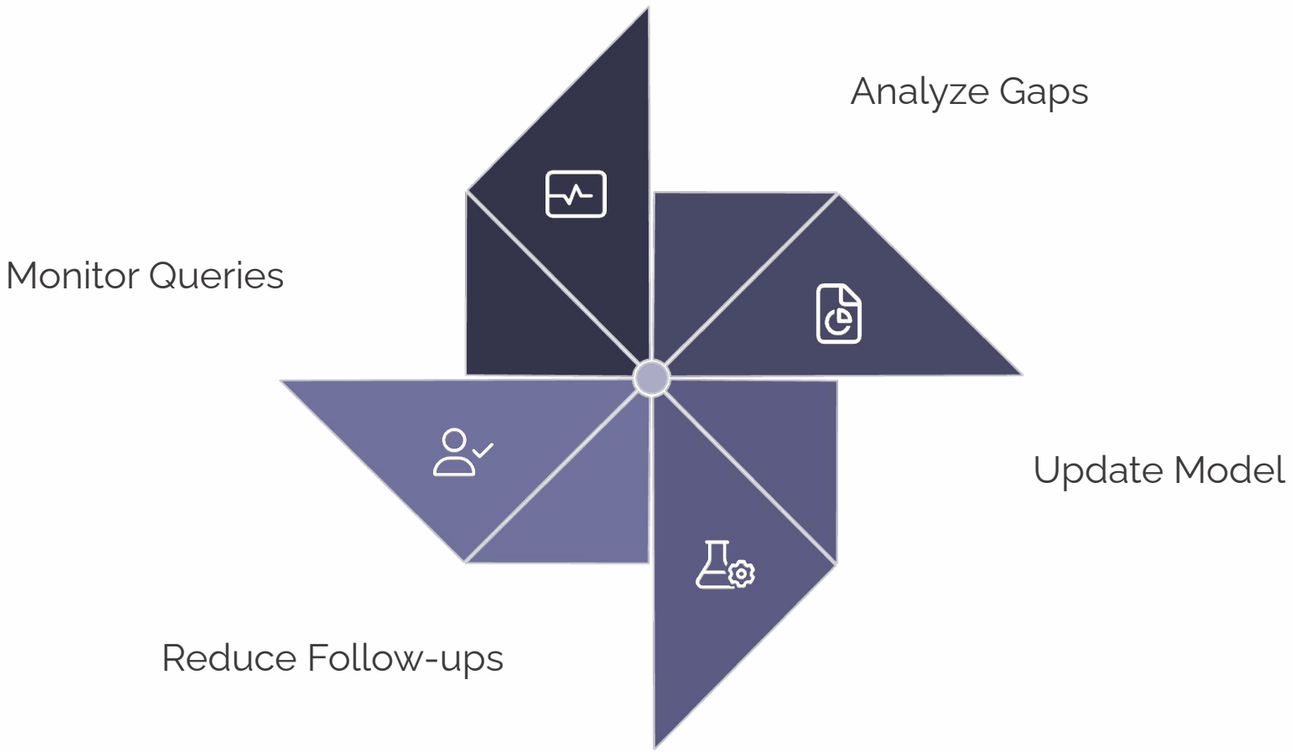
Monitoring supplemental queries can drive improvements to the model for more accurate and detailed responses, reducing the need for follow-up interactions.

Faster User Solutions

Reducing the number of queries leads to faster solutions for users and lower operational costs across the entire system.

Key Cost Metric

One key metric: **cost per successfully completed task** or per retrieved document by the system — a direct measure of operational efficiency.



This continuous improvement cycle ensures that query volume decreases over time as the model becomes more accurate and responsive to user needs.

Token Usage and Cost Monitoring

AI systems cost money to operate — monitoring helps the organization understand what drives those costs and how to control them effectively.

Token Restrictions

If logs indicate high usage of access tokens, implement restrictions on the number of tokens available to users to control runaway costs.

Billing Model Selection

Monitor the number of requests and specific prompt usage to ensure the best configuration and billing model is selected for the organization's needs.

Workload and Hardware Configuration

Each query and response causes a workload on the system — changes in volume can increase or decrease costs significantly.

Scale Up for Heavy Use

Heavily used systems may benefit from an increase in compute, storage, and connectivity configuration — e.g., better CPU/GPU and storage hardware to reduce latency.

High-Performance Trade-off

High-performance hardware reduces latency and processing time but costs more. The investment must be justified by usage volume and performance requirements.

Scale Down for Low Use

On low-use systems, reduce hardware requirements to save money long-term. Right-sizing infrastructure is key to cost-efficient AI operations.

Data Storage and Retention Policies

Why Storage Matters

Storage of prompts, queries, and outputs used in subsequent prompts speeds up processing but costs money. Careful policies are required to balance performance with cost.

LRU Eviction Policy

The **Least Recently Used (LRU)** eviction policy removes the oldest and outdated embeddings in system memory, cache, or vector database when the system reaches predefined limits.

01

Define Retention Policies

Establish when data should be evicted from the record balancing speed benefits against storage costs.

02

Implement Deduplication

Deduplication practices should be documented to reduce redundant storage and lower overall costs.

03

Apply Data Eviction

Data eviction practices remove stale data from memory, cache, and vector databases at predefined thresholds.

04

Monitor Storage Costs

Continuously track storage utilization to ensure policies are working and costs remain within budget.

Rebuild Frequency and Retraining

1. **Document Rebuild Frequency**

   Rebuild frequency should be documented to allow the system to be rebuilt and increase performance and relevancy over time.

2. **Retrain on Newer Datasets**

   Retraining may include updating the model on newer or more accurate datasets to keep it current and aligned with real-world information.

3. **Reduce Hallucinations and Bias**

   Retraining reduces hallucinations and bias within the model's operation, improving the reliability and trustworthiness of AI outputs.

Why Retraining Matters

Without regular retraining, AI models can drift from accuracy — producing outdated, biased, or hallucinated responses. Scheduled rebuilds are a critical part of responsible AI operations.

Cost-Benefit Analysis

Performing a cost-benefit analysis helps evaluate how changes to system configuration affect performance and cost overall. The system budget should be enforced at all times through careful automated monitoring.

By Project

Break down costs by individual project to understand which initiatives consume the most resources and deliver the most value.

By Customer

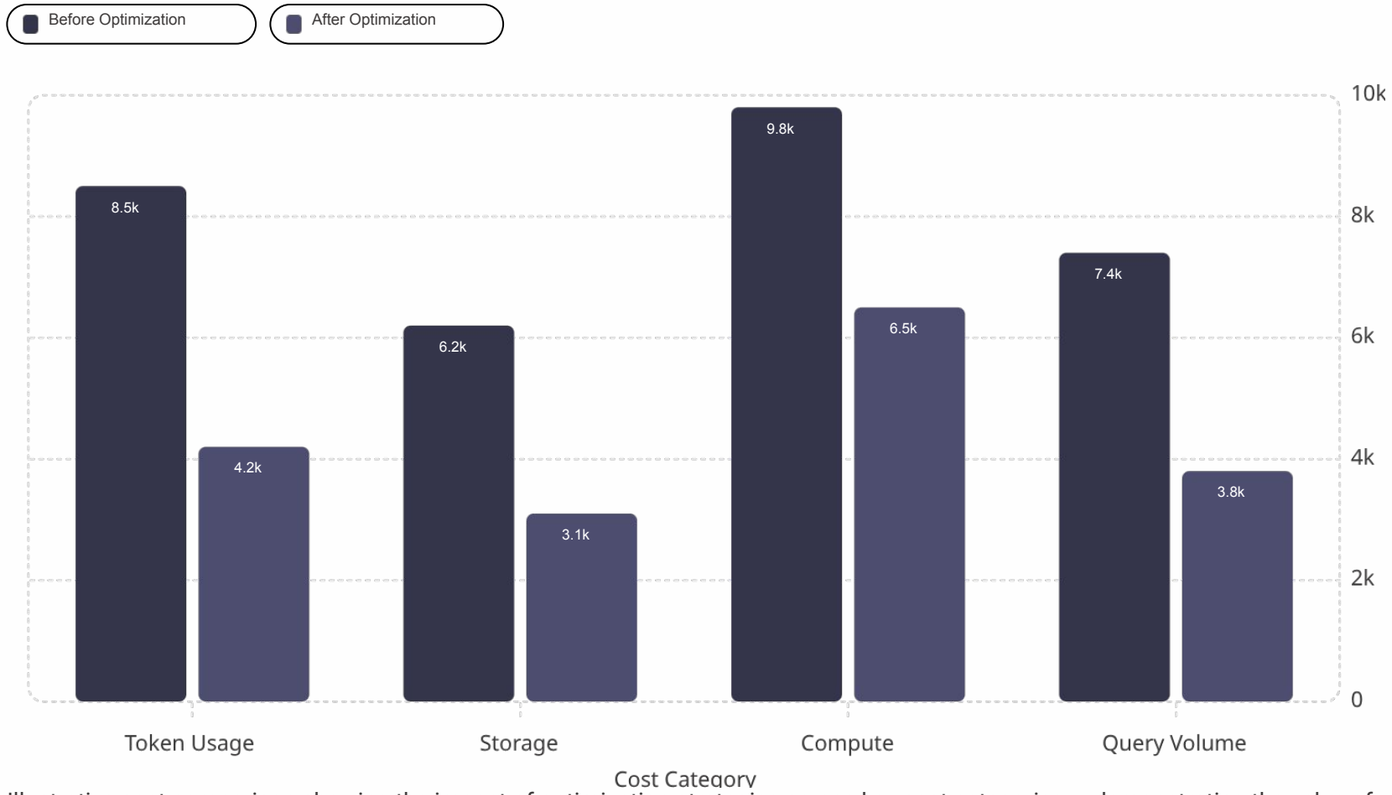
Attribute costs to specific customers or user groups to enable accurate billing and identify high-cost usage patterns.

By Prompt I/O

Compare prompt input and response costs against system computing costs for fine-grained operational insight.

Automated Monitoring

Fine-grained analysis ensures the organization maintains its established budget when operating an AI system at scale.



Illustrative cost comparison showing the impact of optimization strategies across key cost categories — demonstrating the value of continuous cost-benefit analysis.

Key Takeaways

SUMMARY

01

Performance Monitoring

Ensures AI systems meet workload demand and operate efficiently — tracking both prompts (user inputs) and queries (backend requests).

02

Prompt Compression

Lowers cost and speeds up inference in token-limited environments by reducing complexity while preserving meaning.

03

Query & Token Monitoring

Monitoring supplemental queries improves model quality; token usage logs help enforce limits and select the best billing model.

04

Hardware Right-Sizing

Configuration should match usage — scale up for heavy use, scale down for low use to optimize cost and performance.

05

Storage & Retention

Retention policies, deduplication, and LRU eviction manage storage costs effectively and keep the system lean.

06

Rebuild & Cost Analysis

Document rebuild frequency, retrain on newer data to reduce hallucinations and bias, and use cost-benefit analysis with automated budget monitoring to keep operational costs under control.